COLE SHANKS

Electrical engineering graduate from UBC with hands-on experience in embedded systems and digital hardware — firmware in C and digital design in Verilog (FPGA), and the teleoperation firmware (C++) for an autonomous field robot in a team capstone project. Based in Japan, native English speaker with functional Japanese (JLPT N3 / CEFR B1), looking for embedded or software roles, with a current focus on Rust.

cshanks223@gmail.com / coleshanks.github.io / Kure, Hiroshima, Japan

EDUCATION

BASc Electrical Engineering, 2018–2022 — University of British Columbia

Graduated with distinction · Minor in Commerce
Main degree — digital systems design, microcomputers, signals & systems, control systems, electronics, and a year-long capstone project.

BASc Engineering Transfer Program, 2017–2018 — Capilano University

First-year engineering transfer — calculus, physics, and chemistry foundations before transferring to UBC.

BCom, Rotman Commerce, 2016–2017 — University of Toronto

First year of Rotman Commerce — economics, financial accounting, and marketing fundamentals — before switching into electrical engineering.

EXPERIENCE

Assistant Language Teacher — JET Programme

Aug 2022–Present, Kure, Hiroshima, Japan

- Planned and delivered English lessons, teaching and presenting in front of classes.
- Coached students for English speech contests.
- Took part in school life — sports day, culture festival, and other events.
- Developed working Japanese in a full-time Japanese-language office environment.

Application Engineer (Co-op) — Maxim Integrated

Jun 2020–Dec 2020, Burnaby, BC, Canada

- Soak-tested USB/DisplayPort extension hardware — power-cycle, hot-plug, and multi-day endurance runs with pass/fail and error logging.
- Ran a peripheral compatibility matrix (conference cameras, speakerphones, mics) across multiple host setups.
- Validated links in live video calls and measured throughput and data integrity with iperf3 and PassMark BurnInTest.

PROJECTS

monojiten — command-line monolingual Japanese dictionary, built in Rust (2026, ongoing)

Propbot — Autonomous Data-Collection Robot — ECE capstone: a self-driving robot that collects wireless-signal data (2022)

Simple RISC Machine — a RISC (Reduced Instruction Set Computer) CPU built in Verilog and run on an FPGA (2019)

Reflow Oven Controller — 8051-assembly firmware controlling a solder-reflow oven's temperature profile via LCD and button interface (2019)

SKILLS

Rust · Low-level (C, 8051 assembly) · OOP (C++, C#) · FPGA / HDL (Verilog) · Web (HTML, CSS)
git / GitHub · Linux / Unix (fish) · VS Code / Vim / Visual Studio · PlatformIO
Quartus / ModelSim · MATLAB / Simulink · Oscilloscope / Function Generator

LANGUAGES

English — native
Japanese — JLPT N3 · CEFR B1